



Hypothesis Testing Exercises Solutions

Exercise 6.1.

For each of the experiments described below, state the null hypothesis and the alternative hypothesis.

- (a) A runner wishes to improve her 100m time, which is normally distributed with mean μ_0 . She trains intensively for a week, and measures her time for running 100m over 5 days.

H_0 : the time taken to run 100m will not change;

H_1 : the time taken to run 100m will decrease.

- (b) A group of 46 individuals with clinical depression were randomly assigned to either treatment as usual, or sessions playing a computer game. Participants performed a neuropsychological assessment which produced a score. The scores were then compared between the groups. Neuropsychological assessment scores are normally distributed, and higher scores on the test correspond to better mental health.

H_0 : the scores of patients in the group playing the computer game will not change compared to those not playing the game;

H_1 : the scores of patients in that group will increase compared to those in the other group.

- (c) A football fan wishes to determine if there is a biased pattern in the times at which yellow cards are given out during a match, or whether they follow a normal distribution. They watch all the games in a Premier League season and record the number of yellow cards given out during 10 minute intervals of each match.

H_0 : the number of yellow cards given in each 10 minute interval is distributed uniformly;

H_1 : the number of yellow cards is consistently higher or lower in certain intervals across all matches

- (d) A computer gamer achieves a score of 40,000 points on a difficult game, which is a new world record. The scores for the game are normally distributed, and the adjudicator for the world record wishes to determine whether or not the player was cheating.

H_0 : the score was obtained without cheating;

H_1 : the player did not obtain the score without cheating.

Exercise 6.2.

- (a) A biologist studying elk measures the weights of elk in the park where they work over the course of a year, to make sure they are getting enough food. The biologist finds that based on a sample of the population, they have a mean weight of 171kg with a standard deviation of 47kg. Their weights are assumed to be normally distributed. An elk is observed in the park, and found to have a weight of 95kg. Test, using the 5% significance level, whether this elk is underweight.

The null hypothesis is that the elk is an acceptable weight, and the alternative hypothesis is that it is underweight. For significance at 5%, our test statistic is 1.64 (and since we are looking at the lower tail, we can use -1.64) so the critical region is -1.64 or less. For an elk weighing 95kg,

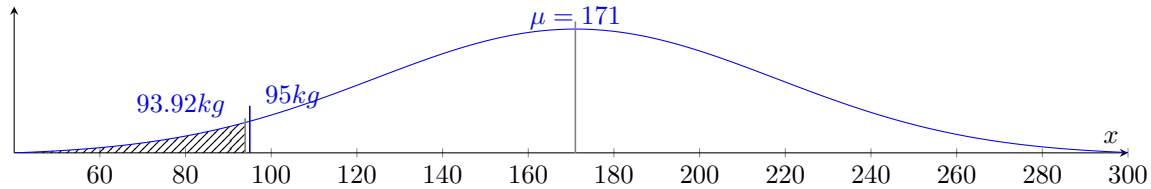
$$z = \frac{x - \mu}{\sigma} = \frac{95 - 171}{47} = -1.61$$

This is outside of the critical region, so we can accept the null hypothesis that the elk is an acceptable weight. For this distribution, the critical value gives a weight of

$$1.64 = \frac{x - 171}{47} \quad \Rightarrow \quad x = (1.64 \times 47) + 171 = 248.08$$

This is the opposite side of the mean, so we would be looking for a minimum weight to accept the null hypothesis of:

$$171 - (248.08 - 171) = 93.92$$



- (b) An ice cream van driver starts a round on a new estate. The mean daily income she receives on this estate during her first month is £46.70 with a standard deviation of £14.30. At the beginning of a second month, she records the daily income from this estate of a random sample of 7 days. The total income for those 7 days is £450. Test, using the 5% significance level, whether the mean daily income from this estate exceeded £46.70. Assume that the sample of daily incomes is from a normal distribution with standard deviation 14.3. If the total income for 7 days was £450, the mean income per day was $\frac{450}{7} = £64.28$. The null hypothesis is that the mean of the incomes is still £46.70, and the alternative hypothesis is that the mean is now greater than £46.70. For 5% significance on an upper tail, the critical region is 1.64 or greater. Using the sampling distribution $X \sim N(46.70, \frac{14.3^2}{7})$, we can calculate the test statistic:

$$z = \frac{x - \mu}{\sigma} = \frac{64.28 - 46.70}{\frac{14.3}{\sqrt{7}}} = 3.25$$

The corresponding probability is $0.0006 = 0.06\%$, which is below the 5% significance level. This result is significant, and we can reject the null hypothesis. There is significant evidence to suggest the mean daily income from the second month is more than £46.70.

Exercise 6.3.

- (a) Research has shown that the mean recovery time for a cold is 18 days, with a standard deviation of 1.5 days, and follows a normal distribution. What are the upper and lower limits for the time taken to recover from a cold, to a significance level of 5%?

We are looking for the values (to the nearest day) corresponding to the critical values, where the test statistic will be 1.96 and -1.96. We can calculate:

$$1.96 = \frac{x - \mu}{\sigma} = \frac{x - 18}{1.5} \quad \Rightarrow \quad x = (1.96 \times 1.5) + 18 = 20.94$$

This means the upper limit is 21 days, and we can find the lower limit by reflecting in the mean:

$$18 - (20.94 - 18) = 15.06$$

So the lower limit is 15 days. We would expect 95% of colds to last between 15 and 21 days.

- (b) At a bakery, a machine is used to measure out quantities of dough for making loaves. For each batch, the mean amount of dough per load is set using a control on the machine. The weights of the resulting loaves are normally distributed with standard deviation 24 grams.

The control is set to produce a batch of loaves with a mean weight of 1000 grams. The weights, in grams, of a random sample of nine loaves from this batch had a mean of 970.1 grams. Using the 5% significance

level, examine whether the mean weight of this batch of loaves is different from 1000 grams.

Use the sampling distribution of the mean, which is given by $\bar{X} \sim N(1000, \frac{24^2}{9})$. Then, using the two-tailed critical value of 1.96 the critical region is anything between -1.96 and 1.96, we can find the z -score as:

$$z = \frac{970.1 - 1000}{\frac{24}{3}} = -3.74$$

This is outside of the critical region, so we can reject the null hypothesis. There is significant evidence to suggest that the mean weight of this batch of loaves is not 1000g. The corresponding probability is $0.00016 < 0.05$.

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- Define your variables
 - Formulate the hypotheses (H_0 and H_1)
 - Declare the test including:
 - which tail/how many tails
 - the type of test
 - the significance level and
 - what you need to carry out the test
 - ...
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Exercise 6.4. For each of the situations described below, follow the list above and suggest what you would write for the first three bullet points.

- (a) A flock of goldfinches eat seeds from a birdfeeder, and the amount of seed each bird consumes is normally distributed. A bird visits the feeder while nobody is watching, and you wish to determine if it was a goldfinch, or one of the larger birds sometimes seen in the area.

- Variables: the amount of seed consumed by the birds
- H_0 : the bird was a goldfinch; H_1 : the bird was another, larger type of bird that eats more seed
- Use:
 - One-tailed z -test (upper tail) with 5% significance
 - Need the mean/SD of the existing normal distribution for X , and the amount of seed eaten by this bird

- (b) The heights of 10-year-old boys are normally distributed. A researcher wants to determine if a new experimental drug some boys were given as babies has affected their height.

- Variables: height of the boys in the study, grouped into those who took the drug and those who didn't
- H_0 : the drug had no impact on height; H_1 : taking the drug has changed the height significantly
- Use:
 - Two-tailed z -test with 5% significance
 - Need the mean/SD of the distribution of boys who didn't take the drug, and the mean of the distribution of those who did; use the sampling distribution of the means of the boys who didn't take it, to see if the mean of the other set has a p -value more or less than the significance level relative to that distribution